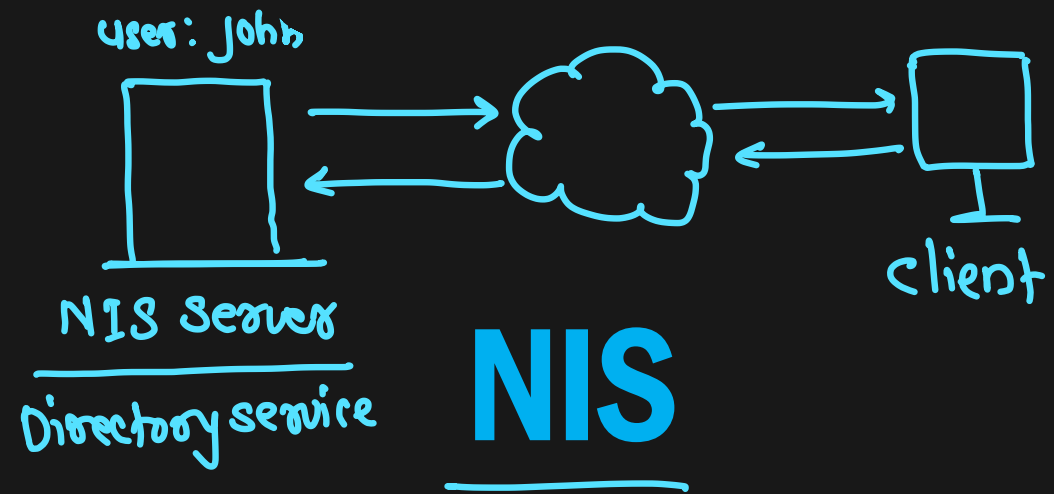






- NIS (Network Information Service) is a client-server directory service protocol developed by Sun Microsystems for distributing system configuration data across a network
- It's often called "YP" (Yellow Pages) - though this name couldn't be trademarked
- Primary purpose is a centralized management of:
 - ✓ ▪ User accounts (/etc/passwd)
 - ✓ ▪ Passwords (/etc/shadow)
 - ✓ ▪ Groups (/etc/group)
 - ✓ ▪ Hostnames (/etc/hosts)
 - ✓ ▪ Network services (/etc/services)
 - ✓ ▪ Network protocols (/etc/protocols)

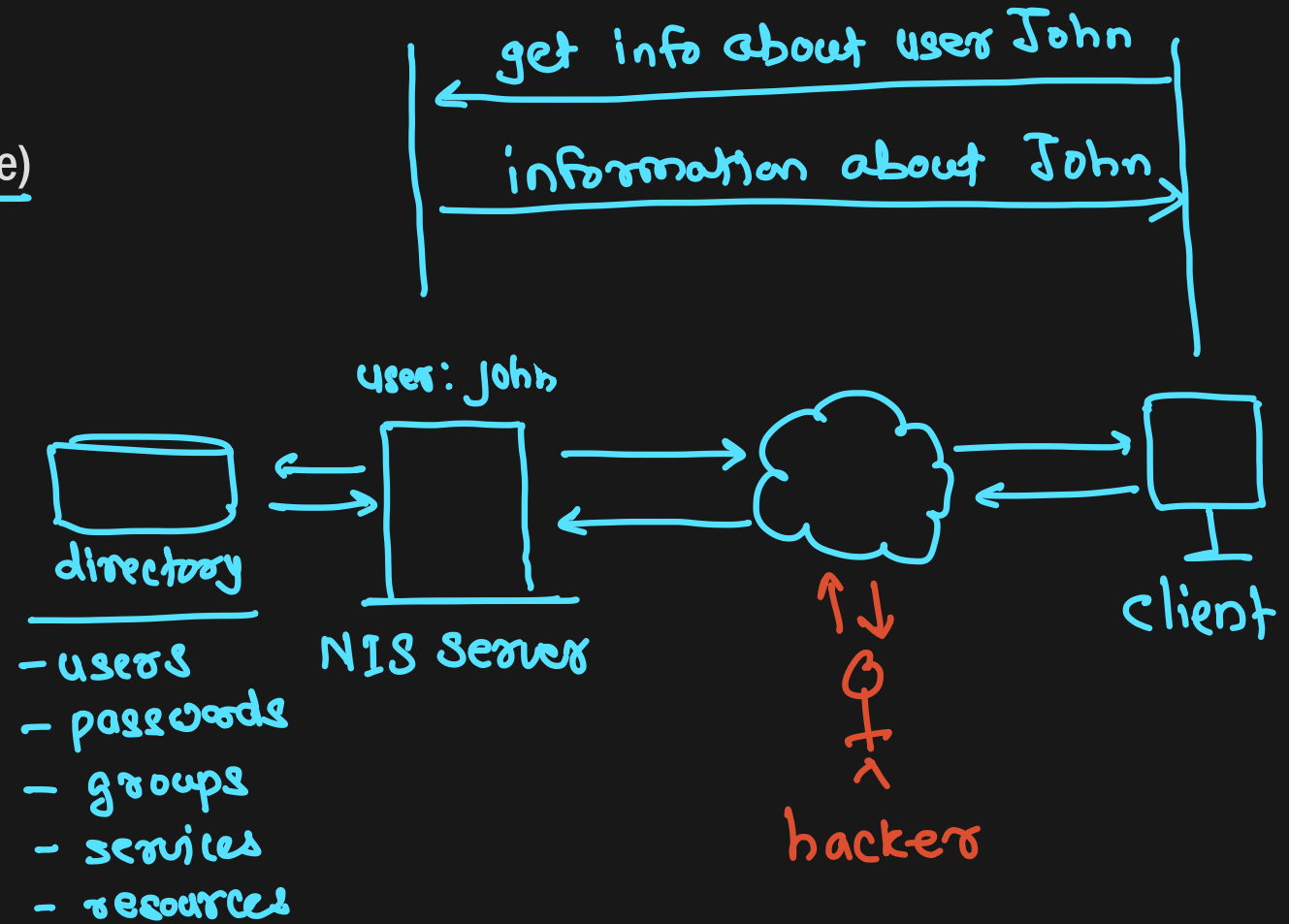
History



- 1985: Sun Microsystems develops NIS (originally "Yellow Pages")
- 1990: Becomes industry standard for Unix networks
- 1992: Name changed from "Yellow Pages" to NIS (trademark issues with British Telecom)
- 1990: NIS+ introduced (enhanced security, more complex)
- 2000: LDAP begins replacing NIS in many environments
- Present: Still used in legacy Unix environments and some clusters

How NIS Works

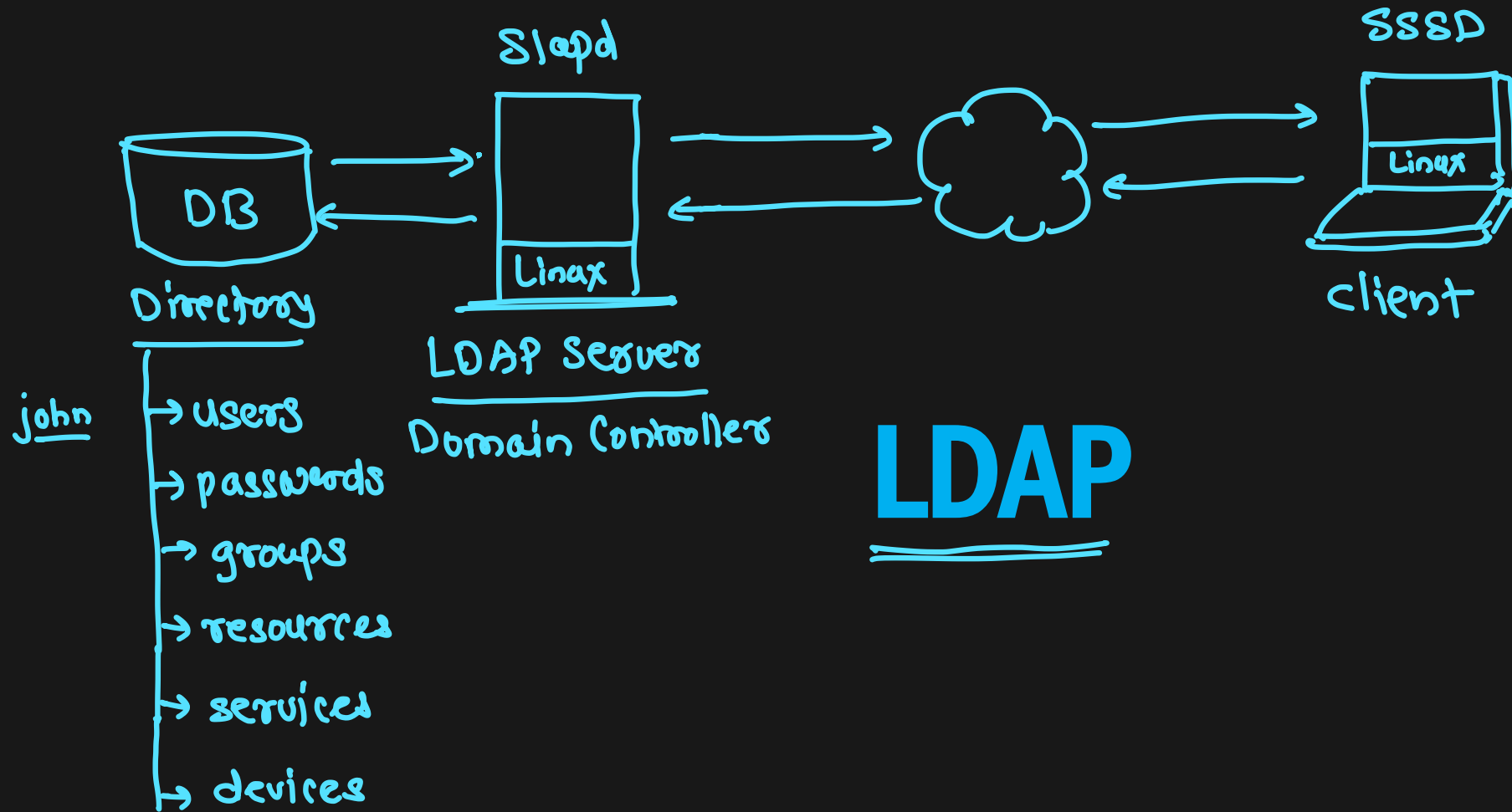
- Client makes a request (e.g., "get user john's info")
- Client's `ypbind` connects to NIS server
- Server receives request
- Server searches appropriate map (passwd.byname)
- Server returns result to client
- Client caches result (optional)





Security Limitations of NIS

- No Encryption - All data (including passwords) transmitted in clear text
- Weak Authentication - Only IP-based restrictions available
- No Password Policies - No password expiration, complexity, or history
- Vulnerable to Spoofing - Clients can't verify server authenticity
- Predictable Ports - Uses well-known ports (111, 953)
- No Access Control - All NIS users can access all maps (except shadow)



LDAP



→ Database

- LDAP is an open, vendor-neutral application protocol for accessing and maintaining distributed directory information services over an IP network
- Think of it as a highly optimized, read-heavy database designed specifically for storing and looking up structured information about people, devices, and resources — like a phonebook that every application on your network can query
- It was derived from the X.500 directory standard and standardized in RFC 4511 (version 3, still current)
- The "Lightweight" in the name refers to it being a simplified version of the older DAP (Directory Access Protocol), designed to run over TCP/IP without requiring the OSI networking stack
- **Ports numbers**
 - 389: TCP/UDP - LDAP (plain text)
 - 636: TCP - LDAPS (encrypted) → Microsoft
 - 3268: TCP - Global Catalog (MS AD) → Active Directory
- **Common Implementations**
 - OpenLDAP: De facto open-source standard; slapd daemon
 - Microsoft Active Directory: LDAP v3 + Kerberos + DNS; most common enterprise deployment
 - FreeIPA / 389 Directory Server: Red Hat ecosystem; integrates Kerberos, DNS, CA
 - Apache Directory Server: Java-based; embedded well in Java apps
 - OpenDJ / ForgeRock DS: Enterprise Java, high-performance
 - eDirectory: Novell/NetIQ

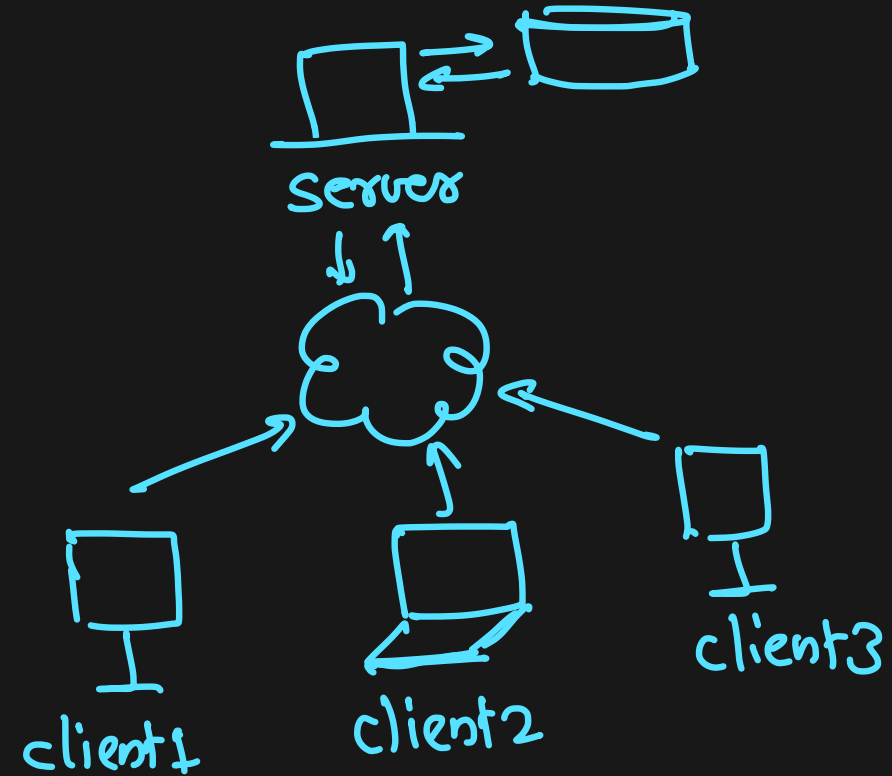
Timeline Summary



Year	Event	Significance
<u>1988</u>	<u>X.500 standard</u>	<u>Original directory service (complex, OSI-based)</u>
<u>1993</u>	<u>LDAP created at U of Michigan</u>	<u>Simplified X.500 for TCP/IP networks</u>
<u>1995-96</u>	<u>LDAPv2 standardized</u>	<u>Independent of X.500, practical deployments</u>
<u>1997-2000</u>	<u>LDAPv3 published</u>	<u>Security, extensions, enterprise-ready</u>
<u>1998</u>	<u>OpenLDAP founded</u>	<u>Free, open-source LDAP server</u>
<u>1999</u>	<u>Active Directory launches</u>	<u>LDAP becomes ubiquitous in business</u>
<u>2003</u>	<u>LDAPv3 becomes IETF Draft Standard</u>	<u>Protocol maturity</u>
<u>2010s</u>	<u>Cloud LDAP services</u>	<u>Modern directory-as-a-service</u>
<u>Present</u>	<u>LDAP still actively used</u>	<u>Foundation for many identity systems</u>

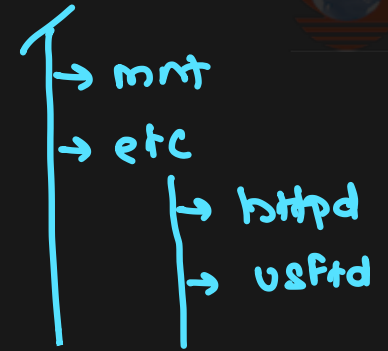
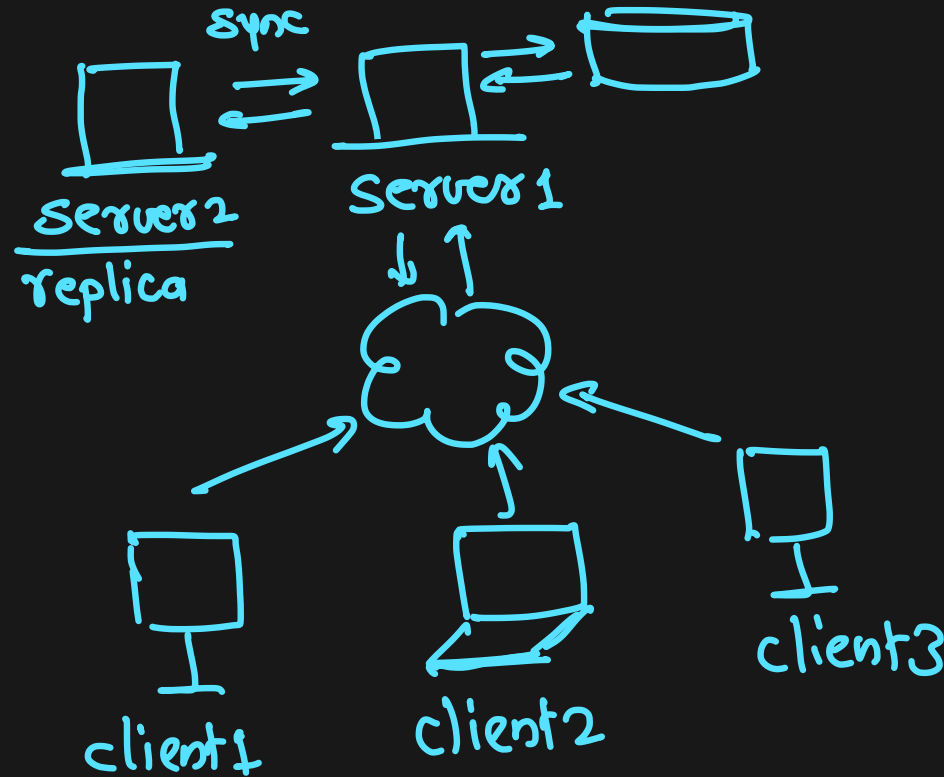
Use cases

- Centralized authentication (users, passwords)
- Address books (email clients)
- Network resource management (printers, servers)
- Single Sign-On (SSO) infrastructure
- Organizational charts and contact information



Key Features of LDAP

- Hierarchical structure (Tree-based)
- Optimized for read-heavy operations
- Centralized authentication
- Platform-independent
- Supports replication
- Secure (via SSL/TLS)



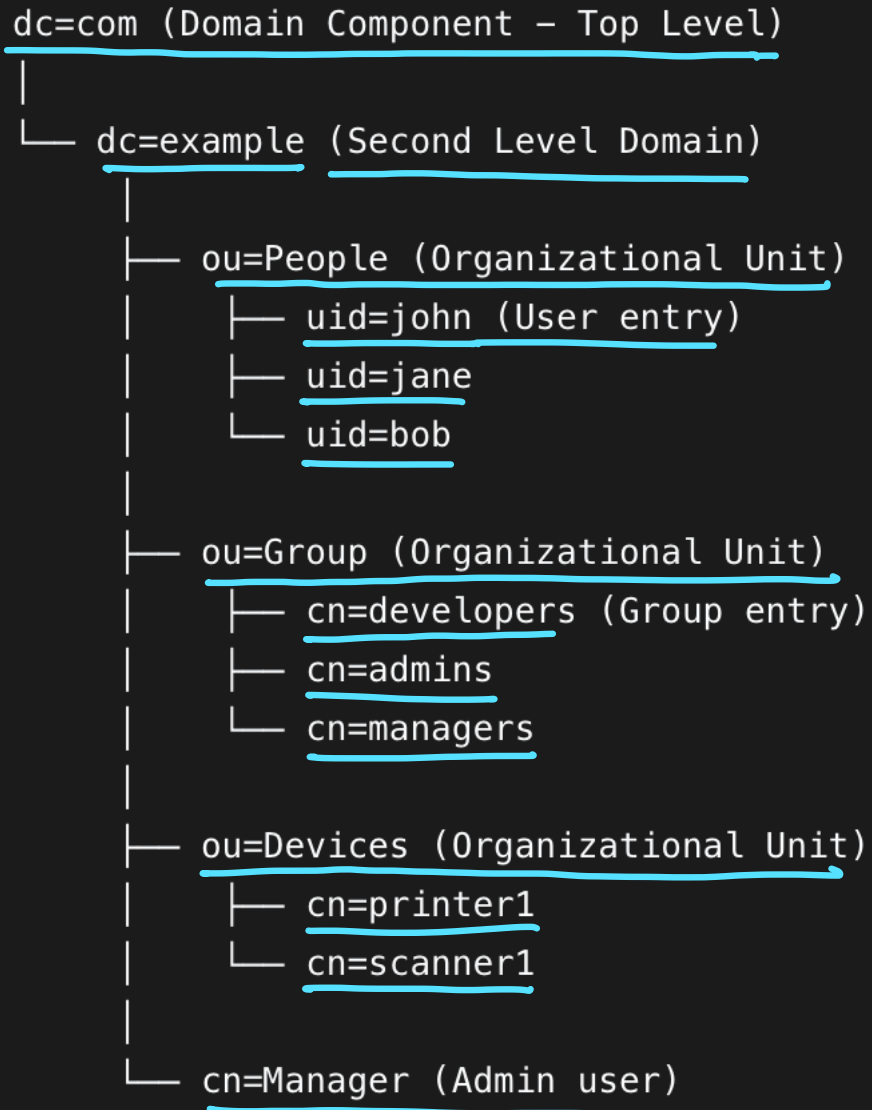
LDAP vs Traditional Database



Feature	LDAP	Relational Database (SQL)
<u>Primary Operation</u>	<u>Read-heavy, search-optimized</u>	<u>Read/Write balanced</u>
<u>Data Structure</u>	<u>Hierarchical tree</u>	<u>Tables with rows/columns</u>
<u>Query Language</u>	<u>LDAP filters (simpler)</u>	<u>SQL (complex)</u>
<u>Transaction Support</u>	<u>Limited</u>	<u>Full ACID compliance</u>
<u>Authentication</u>	<u>Built-in (bind operation)</u>	<u>Application-managed</u>
<u>Typical Use</u>	<u>Identity management, lookups</u>	<u>Transaction processing</u>

- users
- groups
- permissions
- roles
- policies

LDAP Hierarchy Structure



- LDAP stores data in a hierarchical tree structure — not flat tables
- Every entry in the tree has a unique address called a Distinguished Name (DN)
- Example DN:
 - cn=John Doe, ou=Engineering, dc=example, dc=com
- Each comma-separated component is a Relative Distinguished Name (RDN)
 - dc: Domain Component
 - o: Organization
 - ou: Organizational Unit
 - cn: Common Name
 - uid: User ID
 - sn: Surname
 - mail: Email address



LDAP Object Classes

- **An objectClass defines**
 - Which attributes are required (MUST)
 - Which attributes are allowed (MAY)
- **Types of Object Classes**
 - **STRUCTURAL:** Defines the main type of entry (e.g., person, organizationalUnit)
 - **AUXILIARY:** Adds extra attributes (e.g., posixAccount for Unix users)
 - **ABSTRACT:** Template classes (e.g., top - root of all classes)

Object Class	Required Attributes	Optional Attributes	Purpose
<u>top</u>	<u>(none)</u>	(none)	<u>Root class for all entries</u>
<u>person</u>	<u>cn, sn</u>	<u>description, telephoneNumber</u>	<u>Basic person record</u>
<u>organizationalPerson</u>	<u>cn, sn</u>	<u>title, mail</u>	<u>Extended person record</u>
<u>inetOrgPerson</u>	<u>cn, sn</u>	<u>uid, mail, title</u>	<u>Internet/email person</u>
<u>posixAccount</u>	<u>cn, uid, uidNumber, gidNumber, homeDirectory</u>	<u>userPassword, loginShell</u>	<u>Unix user account</u>
<u>posixGroup</u>	<u>cn, gidNumber</u>	<u>userPassword, memberUid</u>	<u>Unix group</u>
<u>organizationalUnit</u>	<u>ou</u>	<u>description</u>	<u>Container for entries</u>

Object class example

```
dn: uid=john,ou=People,dc=example,dc=com
objectClass: top ✓
objectClass: person ✓
objectClass: organizationalPerson
objectClass: inetOrgPerson
objectClass: posixAccount
cn: John Doe
sn: Doe
uid: john
uidNumber: 10001
gidNumber: 10001
homeDirectory: /home/john
userPassword: {SSHA}hashedpassword
mail: john@example.com
title: Software Engineer
```

com (dc=com)

→ example (dc=example)

→ People (ou=People)

uid: john
cn: john doe
sn: doe
mail: john@test.com

entry (cn=john doe)

LDAP Operations



LDAP Operation	CRUD Equivalent	Description
<u>Bind</u>	<u>Authentication</u>	<u>Connect and authenticate to directory</u>
<u>Search</u>	<u>Read</u>	<u>Query for entries matching filters</u>
<u>Add</u>	<u>Create</u>	<u>Add new entry to directory</u>
<u>Modify</u>	<u>Update</u>	<u>Change attributes of existing entry</u>
<u>Delete</u>	<u>Delete</u>	<u>Remove entry from directory</u>
<u>Compare</u>	<u>Check</u>	<u>Verify if entry has specific attribute</u>
<u>Unbind</u>	<u>Disconnect</u>	<u>Close connection</u>

LDAP Search Filters Examples



Find user with uid "john"

(uid=john)

Find all users in People OU

(&(objectClass=posixAccount)(ou=People))

Find users with uid starting with "j"

(uid=j*)

Find users in group "developers" OR "admins"

(|(cn=developers)(cn=admins))

Find users not in specific department

(!(department=IT))

Find users with specific email domain

(mail=*@example.com)

Complex filter: All developers with UID > 10000

(&(cn=developers)(uidNumber>=10000))

LDAP Security



■ Authentication Methods

- Simple Authentication (Basic) → quick check
 - Sends password in clear text (Base64 encoded)
 - Not secure without TLS
 - ldapsearch -x -D "cn=Manager,dc=example,dc=com"
-w password
- SASL (Simple Authentication and Security Layer) → internal
 - Supports GSSAPI (Kerberos), DIGEST-MD5, EXTERNAL
 - More secure than simple bind
- TLS/SSL (LDAPS - Port 636) → production
 - Encrypts entire communication
 - Required for production environments

■ Password Storage Formats

Format	Command	Security Level
✓ <u>{SSHA}</u>	<u>slappasswd -s password</u>	<u>Recommended (SHA-1 + salt)</u>
<u>{SHA}</u>	<u>slappasswd -h {SHA}</u>	<u>Moderate (SHA-1 only)</u>
<u>{CRYPT}</u>	<u>slappasswd -h {CRYPT}</u>	<u>System-dependent</u>
<u>{MD5}</u>	<u>slappasswd -h {MD5}</u>	<u>Weak (not recommended)</u>
<u>{CLEARTEXT}</u>	<u>(plain text)</u>	<u>Never use in production</u>

LDAP Commands



■ Server commands

```
# Start/stop/restart LDAP server
sudo systemctl start slapd
sudo systemctl stop slapd
sudo systemctl restart slapd
```

```
# Check server status
sudo systemctl status slapd
sudo netstat -tulpn | grep :389
```

```
# Backup database
sudo slapcat -l backup.ldif
```

```
# Restore database
sudo slapadd -l backup.ldif
```

■ Client commands

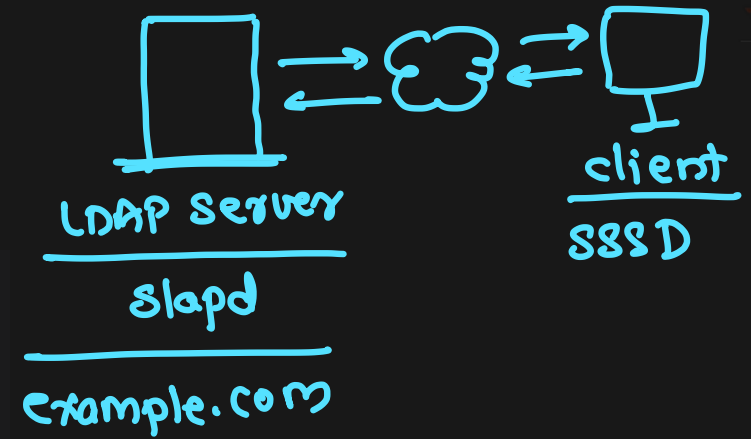
Command	Purpose	Example
<u>ldapsearch</u>	<u>Query directory</u>	<u>ldapsearch -x -b "dc=example,dc=com" "(uid=john)"</u>
<u>ldapadd</u>	<u>Add entries</u>	<u>ldapadd -x -D "cn=Manager" -W -f entry.ldif</u>
<u>ldapmodify</u>	<u>Modify entries</u>	<u>ldapmodify -x -D "cn=Manager" -W -f changes.ldif</u>
<u>ldapdelete</u>	<u>Delete entries</u>	<u>ldapdelete -x -D "cn=Manager" -W "uid=john,ou=People"</u>
<u>ldappasswd</u>	<u>Change password</u>	<u>ldappasswd -x -D "uid=john" -W -S</u>

SSSD (System Security Services Daemon)

- SSSD connects your Linux client to LDAP (and other identity providers)
- **SSSD Configuration** (/etc/sssd/sssd.conf)

```
[sssd]
services = nss, pam           # Name Service Switch + Pluggable Auth Modules
domains = example.com         # Your LDAP domain

[domain/example.com]
id_provider = ldap            # Get user/group IDs from LDAP
auth_provider = ldap          # Authenticate against LDAP
chpass_provider = ldap        # Allow password changes
ldap_uri = ldap://172.16.140.216/
ldap_search_base = dc=example,dc=com
ldap_id_use_start_tls = False  # Disable TLS for lab (use True in production)
cache_credentials = True      # Cache credentials for offline use
enumerate = True              # List all users (set False for large directories)
```



LDAP Schema Files



- Schemas define the rules for directory data structure.
- **Location:** /etc/openldap/schema/
- **Important files:**
 - core.ldif: Basic LDAP objects
 - cosine.ldif: Cosine Internet X.500 schema
 - inetorgperson.ldif: Internet organizational person
 - nis.ldif: Network Information Service (Unix)
 - openldap.ldif: OpenLDAP-specific objects
- **Importing Schemas**
 - sudo ldapadd -Y EXTERNAL -H ldapi:/// -f /etc/openldap/schema/cosine.ldif

↓
SASL

→ add information in ldap server

LDAP Utilities & Tools



■ Command Line

- Idapsearch: Query directory
- Idapvi: Edit LDAP entries with text editor
- slapcat: Export database to LDIF
- slapindex: Re-index database
- slappasswd: Generate password hashes ✓

■ GUI Tools

- Apache Directory Studio (cross-platform)
- phpLDAPadmin (web-based)
- LDAP Admin (Windows)
- LDAP Account Manager (web-based)

LDIF (LDAP Data Interchange Format)



```
dn: uid=jsmith,ou=People,dc=example,dc=com
objectClass: top
objectClass: person
objectClass: organizationalPerson
objectClass: inetOrgPerson
objectClass: posixAccount
cn: John Smith
sn: Smith
uid: jsmith
uidNumber: 10002
gidNumber: 10002
homeDirectory: /home/jsmith
userPassword: {SSHA}hashedpassword
mail: jsmith@example.com
title: System Administrator
telephoneNumber: +1 555-123-4567
```

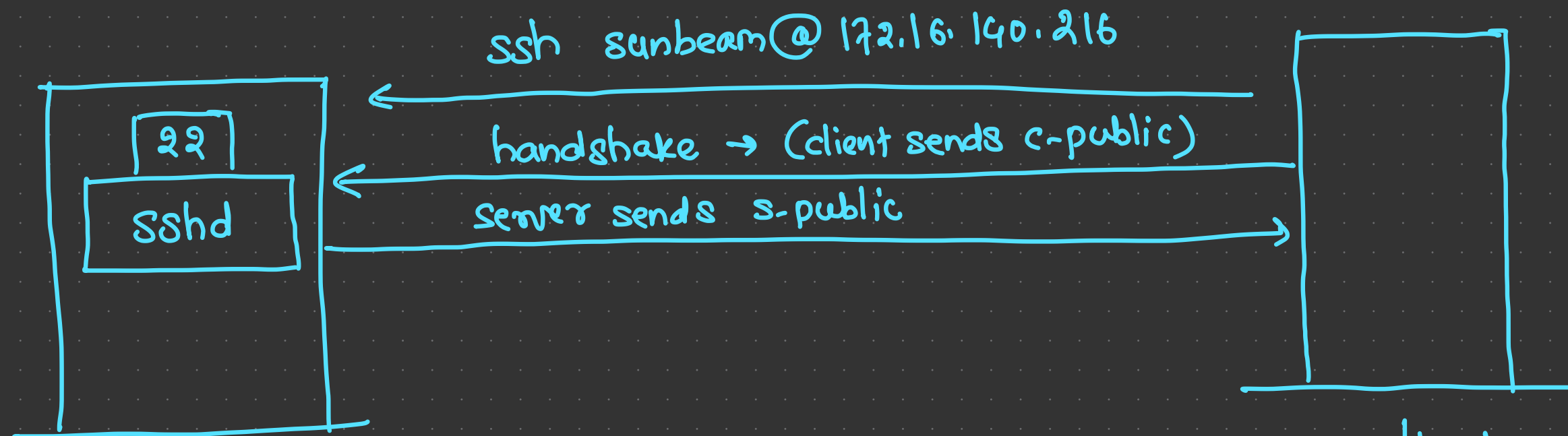
NIS vs LDAP



deprecated

preferred

Feature	NIS	LDAP
<u>Year Introduced</u>	<u>1985</u>	<u>1993</u>
<u>Security</u>	<u>Very weak (no encryption)</u>	<u>Strong (TLS, SASL)</u>
<u>Authentication</u>	<u>Simple password auth</u>	<u>Bind with various mechanisms</u>
<u>Data Model</u>	<u>Flat maps</u>	<u>Hierarchical tree</u>
<u>Scalability</u>	<u>Limited (flat namespace)</u>	<u>Excellent (hierarchical)</u>
<u>Replication</u>	<u>Simple master-slave</u>	<u>Complex multi-master</u>
<u>Schema</u>	<u>Fixed</u>	<u>Extensible</u>
<u>Standard</u>	<u>Proprietary (Sun)</u>	<u>IETF Standard</u>
<u>Modern Support</u>	<u>Legacy systems</u>	<u>All modern systems</u>
<u>Complexity</u>	<u>Simple</u>	<u>Moderate</u>
<u>Query Capability</u>	<u>Basic key lookup</u>	<u>Complex searches/filters</u>



SSH server
172.16.140.216

SSH

s-public
encryption

s-private
decryption

c-public
encryption

c-private
decryption

